United States Patent	12418237
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lu; Jiangheng et al.

Power system with adaptive minimum frequency

Abstract

A power system with adaptive minimum frequency is disclosed. The power system includes a resonant converter and a control circuit. Under the control of the control circuit, the resonant converter works with an adaptive minimum frequency. The value of the adaptive minimum frequency is preset and is selected by an equivalent resistance or an output voltage of the resonant converter.

Inventors: **Lu; Jiangheng (Livermore, CA), Wang; Siran (Hangzhou, CN)**

Applicant: **Monolithic Power Systems, Inc.** (Kirkland, WA)

Family ID: **1000008820722**

Assignee: **Monolithic Power Systems, Inc. (Kirkland, WA)**

Appl. No.: **18/071853**

Filed: **November 30, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240178744 A1	May. 30, 2024

Publication Classification

Int. Cl.: **H02M3/00** (20060101); **H02M1/00** (20070101); **H02M3/335** (20060101)

U.S. Cl.:

CPC **H02M3/015** (20210501); **H02M1/0058** (20210501); **H02M3/01** (20210501); **H02M3/33571** (20210501);

Field of Classification Search

CPC: H02M (1/0058); H02M (1/083); H02M (3/01); H02M (3/015)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8238123	12/2011	Zhang	363/21.13	H02M 3/338
8687387	12/2013	Li	363/21.13	H02M 3/156
10263528	12/2018	Ouyang	N/A	N/A
11051377	12/2020	Xiong	N/A	H05B 47/25
2013/0221758	12/2012	Kai	307/104	H02J 50/10
2014/0085938	12/2013	Shi	363/21.01	H02M 3/33523
2014/0211515	12/2013	Tomioka	363/21.02	H02M 3/33571
2015/0049515	12/2014	Zhao	363/17	H02M 3/33507
2016/0190945	12/2015	Liu	363/21.02	H02M 3/3376
2016/0241145	12/2015	Matsuura	N/A	H02M 1/14
2017/0012541	12/2016	Ye	N/A	H02M 1/4258
2017/0093296	12/2016	Chen	N/A	H02M 1/088
2018/0019677	12/2017	Chung	N/A	H03J 1/0091
2020/0067407	12/2019	Kashiwagi	N/A	H02M 3/156
2022/0385193	12/2021	Lai	N/A	H02M 7/5387

Primary Examiner: Rosario-Benitez; Gustavo A

Attorney, Agent or Firm: Perkins Coie LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to electronic circuits, and more particularly but not exclusively to power systems with resonant converters.

BACKGROUND

(2) A resonant circuit enables a resonant converter to change its gain (e.g., to compensate for changes at its input and/or the requirements of a load) by adjusting switching frequency of power switches of the resonant converter. The resonant converter may operate the power switches over a wide range of switching frequencies to achieve the required output voltage or current at the required time. The gain of the resonant converter over a wide range of switching frequencies for different values of quality factor Q is plotted as shown in FIG. **1**. It can be seen that each gain curve has a peak which defines the boundary between an inductive and a capacitive impedances of the resonant converter, hence an inductive operation region and a capacitive operation region are defined as shown in FIG. **1**, wherein the left of the dotted line is the capacitive operation region, and the right of the dotted line is the inductive operation region. The objective of defining both regions is to maintain an inductive operation across an entire input voltage and output voltage/current ranges, and never fall into the capacitive operation region. Such requirement is due to that Zero Voltage Switching (ZVS) for saving the switching loss is only achieved in the

inductive operation region.
(3) As can be seen from FIG. **1**, the peaks of the different gain curves are located at different switching frequencies. To make sure the resonant converter works in the inductive operation region under all input and output conditions, a minimum switching frequency is set. However, with the minimum switching frequency limit, a maximum available output voltage on the gain curve with smaller Q is limited. To expand the output voltage range for gain curve with small Q, higher input voltage are required, and may lead to higher requirement for the devices of the resonant converter and increase the cost.
SUMMARY
(4) It is an object of the present invention to set an adaptive minimum switching frequency for a power system with a resonant converter to achieve a wide input/output range, with reduced current and voltage stress to the devices in a resonant converter.
(5) In accomplishing the above and other objects, there has been provided, in accordance with an embodiment of the present invention, a power system comprising: a resonant converter, configured to convert an input voltage to an output voltage to power a load; and a control circuit, configured to provide a drive control signal for controlling the resonant converter working with a frequency having a minimum value varied based on the output voltage.
(6) In accomplishing the above and other objects, there has been provided, in accordance with an embodiment of the present invention, a control circuit of a resonant converter, wherein the resonant converter provides an output voltage and an output current to power a load, comprising: a minimum frequency circuit, configured to provide a minimum frequency signal based on the output voltage of the resonant converter; and a control unit, configured to provide a drive control signal for controlling the resonant converter working with a frequency having a minimum value determined by the minimum frequency signal.
(7) In accomplishing the above and other objects, there has been provided, in accordance with an embodiment of the present invention, a control method for controlling a resonant converter, wherein the resonant converter works with a frequency and provides an output voltage and an output current to a load, the control method comprising: presetting at least two minimum values for a lower limit of the frequency of the resonant converter; and selecting one of the at least two minimum values as the lower limit of the frequency of the resonant converter based on the output voltage of the resonant converter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows prior art gain curves of a resonant converter with different quality factors.
(2) FIG. **2** schematically shows a power system **20** in accordance with an embodiment of the present invention.
(3) FIG. **3** schematically shows a power system **30** in accordance with an embodiment of the present invention.
(4) FIG. **4** schematically shows the gains of the power system **30** over a wide range of the switching frequencies with different equivalent resistance in accordance with an embodiment of the present invention.
(5) FIG. **5** schematically shows a frequency generating circuit **50** in accordance with an embodiment of the present invention.
(6) FIG. **6** schematically shows a power system **60** in accordance with an embodiment of the present invention.
(7) FIG. **7** shows a control method **70** for controlling a resonant converter in accordance with an embodiment of the present invention.

(8) The use of the same reference label in different drawings indicates the same or like components.

DETAILED DESCRIPTION

(9) In the present invention, numerous specific details are provided, such as examples of circuits, components, and methods, to provide a thorough understanding of embodiments of the invention. Persons of ordinary skill in the art will recognize, however, that the invention can be practiced without one or more of the specific details. In other instances, well-known details are not shown or described to avoid obscuring aspects of the invention.

(10) FIG. **2** schematically shows a power system **20** in accordance with an embodiment of the present invention. As shown in FIG. **2**, the power system **20** includes a power source **201**, a resonant converter **202**, a load **203** and a control circuit **21** for controlling the resonant converter **202**. The control circuit **21** includes a minimum frequency circuit **204**, a control unit **205** and a drive circuit **206**. The resonant converter **202**, the load **203**, the minimum frequency circuit **204**, the control unit **205** and the drive circuit **206** may be independent or may represent any combination of one or more components that provide the functionality of the power system **20** as described herein.

(11) The power source **201** provides electrical energy, in the form of power, at link **111**. Numerous examples of the power source **201** exist and may include, but are not limited to, power grids, generators, power transformers, batteries, or any other form of electrical power devices capable of providing electrical power to the power system **20**. As referred to herein, the voltage that the power source **201** provides is "the input voltage Vin" of the power system **20**.

(12) The load **203** receives, via a link **112**, electrical power (e.g., voltage, current, etc.) provided by the power source **201** and converted by the resonant converter **202**. Numerous examples of the load **203** exist and may include, but are not limited to, computing devices and related components, such as microprocessors, electrical components, circuits, laptops, desktop computers, mobile phones, or any other type of electrical device and/or circuitry that receives a voltage or a current from a resonant converter.

(13) The resonant converter **202** is a switch-based power converter that converts the electrical energy provided by the power source **201** into a usable form of electrical power required by the load **203** by relying, during at least some of its switching cycles, on zero voltage switching across one or several power switches. Examples of the resonant converter **202** include any type of LLC converter, LCC converter, CLLC converter, CLLLC converter or the like.

(14) Together, the minimum frequency circuit **204**, the control unit **205** and the drive circuit **206** control the resonant converter **202** to vary the amount of power provided to the load **203**. The control unit **205** may be coupled to the drive circuit **206** via link **116** to send drive control signals or commands to the drive circuit **206** for controlling the operations of the resonant converter **202**. For example, the control unit **205** may vary the drive control signals sent to the drive circuit **206** so as to vary the switching frequency of the resonant converter **202** to increase or decrease the output voltage Vout provided to the load **203**. The control unit **205** is coupled to the load **203** via link **113**, and the minimum frequency circuit **204** is coupled to the load **203** via link **114**. Each of the control unit **205** and the minimum frequency circuit **204** receives required information indicative of the various electrical characteristics (e.g., voltage levels, current levels, etc.) associated with the load **203**. The minimum frequency circuit **204** receives the load information, and provides a minimum frequency signal to the control unit **205** via the link **115**. In some embodiments, the control unit **205** may also be coupled to the power source **201** via link **118** to monitor the power source **201** and to control the corresponding operations.

(15) The control unit **205** may comprise any suitable arrangement of hardware, software, firmware, or any combination thereof, to perform the techniques attributed to the control unit **205** herein. For example, the control unit **205** may include switching control circuits with a current loop, a voltage loop or a combination of both. Also, the control unit may be implemented by digital solution, and may include any one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other

equivalent integrated or discrete logic circuitry, as well as any combination of such components. When the control unit **205** includes software or firmware, the control unit **205** further includes any necessary hardware for storing and executing the software or firmware, such as one or more processors or processing units.

(16) The control unit **205** may provide one or more drive control signals across the link **116** that the drive circuit **206** uses to generate one or more gate control signals that cause the power switches of the resonant converter **202** to be on and off.

(17) In some embodiments, the control unit **205** and the drive circuit **206** may together, vary the amount of power that passes from the power source **201** to the load **203**, by varying a duty cycle and/or a switching frequency of the gate control signal(s) provided via the link **117**. The gate control signal(s) are generated in response to the drive control signal(s) provided by the control unit **205** based on the information received from the load **203** and/or the power source **201**. In this way, the control unit **205** and the drive circuit **206** together control the power transfer between the power source **201** and the load **203**.

(18) The links **111**, **112**, **113**, **114**, **115**, **116**, **117** and **118** electrically couple the components of the power system **20**. Each of the links **111**, **112**, **113**, **114**, **115**, **116**, **117** and **118** represent any wired or wireless medium capable of conducting electrical power or electrical signals from one location to another. Examples of the links **111**, **112**, **113**, **114**, **115**, **116**, **117** and **118** include, but are not limited to, physical and/or wireless electrical transmission mediums such as electrical wires, electrical traces, and the like.

(19) FIG. **3** schematically shows a power system **30** in accordance with an embodiment of the present invention. As shown in FIG. **3**, the power system **30** includes a power source **301**, a resonant converter **302**, a minimum frequency circuit **304**, a control unit **305** and a drive circuit **306** which may be independent or may represent any combination of one or more components that provide the functionality of the power system **30** as described herein.

(20) The resonant converter **302** has an input terminal configured to receive an input voltage Vin from the power source **301**, and an output terminal configured to provide an output voltage Vout and an output current Iout to a load, like the load **203** in FIG. **2**. The resonant converter **302** includes: a high side power switch P**1** and a low side power switch P**2** coupled in a half-bridge way, a transformer T**1** having a primary winding Lp, a first secondary winding Ls**1** and a second secondary winding Ls**2**, a resonant capacitor Cr, a first diode D**1** and a second diode D**2**. The primary winding Lp is coupled in series with the resonant capacitor Cr between a switching node SW and a primary ground reference PGND, wherein the switching node SW is a connection node of the high side power switch P**1** and a low side power switch P**2**. A resonant inductor Lr shown in FIG. **3** could be a discrete component couple in series with the primary winding Lp and the capacitor Cr, or could be a symbol representing leakage inductance of the transformer T**1**. The first diode D**1** is coupled in series with the first secondary winding Ls**1** to form a current loop for a current flowing through the first secondary winding Ls**1**, and the second diode D**2** is coupled in series with the second secondary winding Ls**2** to form a current loop for a current flowing through the second secondary winding Ls**2**. The current loop with the first diode D**1** and the current loop with the second diode D**2** alternately provide the output current Iout to the load, and meanwhile charge an output capacitor Cout.

(21) The minimum frequency circuit **304** includes: an equivalent resistance detecting circuit **3041**, a selecting signal generating circuit **3042** and a minimum frequency selecting circuit **3043**. The equivalent resistance detecting circuit **3041** has a first input terminal and a second input terminal coupled to the output terminal of the resonant converter **302** to respectively detect the output voltage Vout and the output current Iout of the resonant converter **302**, and an output terminal configured to provide an equivalent resistance signal Rfb based on the output voltage Vout and the output current Iout. The selecting signal generating circuit **3042** has an input terminal coupled to the output terminal of the equivalent resistance detecting circuit **3041** to receive the equivalent

resistance signal Rfb, and an output terminal configured to provide a selecting signal J**1** based on the equivalent resistance signal Rfb. The minimum frequency selecting circuit **3043** has a control terminal coupled to the output terminal of the selecting signal generating circuit **3042** to receive the selecting signal J**1**, and an output terminal configured to provide a minimum frequency signal fmin based on the selecting signal J**1**, wherein the value of the minimum frequency signal fmin is preset and is selected by the selecting signal J**1**.

(22) In the embodiment of FIG. **3**, the equivalent resistance detecting circuit **3041** includes a feedback detecting circuit **3041**A and a resistance calculating circuit **3041**B. The feedback detecting circuit **3041**A detects the output voltage Vout and the output current Iout, and provides a voltage feedback signal Vfb indicative of the output voltage Vout, and a current feedback signal Ifb indicative of the output current Iout. The resistance calculating circuit **3041**B receives the voltage feedback signal Vfb and the current feedback signal Ifb, and provides the equivalent resistance signal Rfb based on a calculating result of the voltage feedback signal Vfb and the current feedback signal Ifb, specifically, Rfb=Vfb/Ifb=Vout/Iout. It should be known that, the output voltage Vout is represented by the voltage feedback signal Vfb, and the output current Iout is represented by the current feedback signal Ifb, thus the value of the equivalent resistance signal Rfb, which is a quotient of the output voltage Vout and the output current Iout, could be obtained by Vfb/Ifb.

(23) The feedback detecting circuit **3041**A may include any prior art current/voltage sense circuits, and isolating feedback circuits for passing signals between isolated sides, like a transformer or an optocoupler. The resistance calculating circuit **3041**B includes a calculating circuit performing the function of dividing the voltage feedback signal Vfb by the current feedback signal Ifb to generate the equivalent resistance signal Rfb.

(24) The selecting signal generating circuit **3042** may comprise a comparison circuit, which compares the equivalent resistance signal Rfb with at least one resistance threshold signals to provide a comparison result to indicate the value range of the equivalent resistance signal Rfb, so that to provide the selecting signal J**1** accordingly. And then the selecting signal J**1** is provided to the minimum frequency selecting circuit **3043** to select the value of the minimum frequency signal fmin. It should be understood that the selecting signal J**1** is not necessary a single signal, it may present as a set of signals, e.g., the selecting signal J**1** may be a set of the comparison signals based on the comparison result of the equivalent resistance signal Rfb with the at least one resistance threshold signals.

(25) In some embodiments, the selecting signal generating circuit **3042** may be implemented by a digital circuit, like an ADC (Analog-Digital Converting circuit). In that case, the selecting signal J**1** is a digital signal has as many digits as required.

(26) The control unit **305** receives the minimum frequency signal fmin, together with a load information signal Sout representing the output voltage Vout, the output current Iout or both, that the control unit **305** needs, and provides a drive control signal VG based thereon. In the present invention, any control circuit for controlling a half bridge with a frequency limited by the minimum frequency signal fmin as shown in FIG. **3** could be used.

(27) The drive circuit **306** receives the drive control signal VG, and generates a first gate control signal G**1** and a second gate control signal G**2** based on the drive control signal VG. In one embodiment, one of the first gate control signal G**1** and the second gate control signal G**2** has a same phase with the drive control signal VG, while the other one has an opposite phase. Both of the first gate control signal G**1** and the second gate control signal G**2** are enhanced by the drive circuit **306** to have enough drive capability, and then turn on and off the high side power switch P**1** and the low side power switch P**2** alternately.

(28) The drive circuit **306** may include logic circuits to generate the first gate control signal G**1** and the second gate control signal G**2** with opposite phases, and buffers to enhance the driving capability of the first gate control signal G**1** and the second gate control signal G**2**.

(29) FIG. **4** schematically shows the gains of the power system **30** over a wide range of the

switching frequencies with different quality factors, in accordance with an embodiment of the present invention. As can be seen from FIG. **4**, a first minimum frequency f**1** and a second minimum frequency f**2** are preset for the power system **30**. In one embodiment, the selecting signal generating circuit **3042** includes a comparator, the comparator compares the equivalent resistance signal Rfb with a resistance threshold signal Ref. When the equivalent resistance signal Rfb is smaller than the resistance threshold signal Rref, the minimum frequency signal fmin has a value associated with the first minimum frequency f**1**, which makes the drive control signal VG switch the high side power switch P**1** and the low side power switch P**2** with a lower limit defined by the first minimum frequency f**1**. When the equivalent resistance signal Rfb is larger than the resistance threshold signal Rref, the minimum frequency signal fmin has a value associated with the second minimum frequency f**2**, which makes the drive control signal VG switch the high side power switch P**1** and the low side power switch P**2** with a lower limit defined by the second minimum frequency f**2**.

(30) The gain curve GC**1** represents the gain of the power system **30** when working with a lower equivalent resistance, while the gain curve GC**2** represents the gain of the power system **30** when working with a higher equivalent resistance. As can be seen from FIG. **4**, the first minimum frequency f**1** properly divides the inductive operation region and the capacitive operation region for the gain curve GC**1**. However, with the limit of the first minimum frequency f**1**, most part of the inductive operation region of the gain curve GC**2** is wasted, and high output voltage could not be reached when the input current and the input voltage are maintaining unchanged. In the embodiment of FIG. **4**, the second minimum frequency f**2** is set for the gain curve GC**2**. By using a lower minimum frequency limit, i.e., f**2**, the available inductive operation region of the gain curve GC**2** is expanded, i.e., the resonant converter **302** is allowed to extend its switching frequency lower, thus sufficient gain and higher voltage can be reached under the same condition.

(31) The first minimum frequency f**1** and the second minimum frequency f**2** are preset according to the gain curves of the application and are selected based on the equivalent resistance Rfb of the resonant converter **302**, which effectively prevent the resonant converter **302** working in the capacitive operation region.

(32) It should be understood that the embodiment in FIG. **4** with two optional minimum frequencies is for illustration and provides an easy way to understand the present invention. In other embodiments, there may be more optional frequencies to provide a more flexible solution, i.e., a plurality of minimum frequency could be preset, and be selected based on the selecting signal J**1**.

(33) The plurality of minimum frequency is preset according to the specs of the application. For example, persons of ordinary skill in the art could set a frequency corresponds to the peak of the gain curve, or a frequency on the right of the peak (as shown in FIG. **1**) of the gain curve, as the minimum frequency of the corresponding gain curve. Then these minimum frequencies are divided to several groups according to different output voltage ranges, different equivalent resistance ranges, or other specs of the resonant converter. The largest minimum frequency in the group is preset as the minimum frequency for the associated output voltage range or the associated equivalent resistance range. As a result, by the selecting signal J**1**, the minimum frequency is selected for corresponding equivalent resistance or the corresponding output voltage of the resonant converter.

(34) In some embodiments, the minimum frequency selecting circuit **3043** includes a look-up table. The look-up table may have at least two preset frequency values associated with different values of the selecting signal J**1** respectively. The frequency values in the look-up table, and the corresponding relation of the frequency values and the selecting signal J**1**, are determined by specs of the application, like the inductance of the transformer T**1** and the capacitance of the resonant capacitor Cr, etc. In one embodiment, the look-up table could be set by users via a digital interface to the minimum frequency circuit **304**.

(35) In other embodiments, the minimum frequency selecting circuit **3043** may include other circuits which could provide the minimum frequency signal fmin with different values in response to the selecting signal J**1**.
(36) FIG. **5** schematically shows a frequency generating circuit **50** in accordance with an embodiment of the present invention. As shown in FIG. **5**, the frequency generating circuit **50** includes a minimum frequency selecting circuit **501** and a current-frequency converting circuit **502**. In one embodiment, when the frequency generating circuit **50** is utilized with the power system **30** in FIG. **3**, the current-frequency converting circuit **502** may be utilized as part of the control unit **305**.
(37) The minimum frequency selecting circuit **501** has a first terminal coupled to the current-frequency converting circuit **502**, a second terminal connected to the primary ground reference PGND, and a control terminal configured to receive the selecting signal J**1**. The minimum frequency selecting circuit **501** presents as a resistor with an equivalent resistance Rfmin determined by the selecting signal J**1**. The minimum frequency selecting circuit **501** may include a resistor matric behaving like a variable resistor controlled by the selecting signal J**1**.
(38) The current-frequency converting circuit **502** has a first input terminal coupled to the minimum frequency selecting circuit **501** to receive the minimum frequency signal fmin, a second input terminal configured to receive the load information signal Sout, and an output terminal configured to provide a frequency signal Fr based on the minimum frequency signal fmin and the load information signal Sout. In FIG. **5**, the minimum frequency signal fmin presents in the form of a current Ia flowing through the minimum frequency selecting circuit **501**. The load information signal Sout may be the output voltage Vout, the output current Iout, other load indicating signals of the power systems of the present invention or any combination of one or more signals indicating the load information.
(39) In the embodiment of FIG. **5**, the current-frequency converting circuit **502** includes a current control circuit **502**A, an oscillating circuit **502**B. The oscillating circuit **502**B includes a charging current source Ic, a discharging current source Id, a charge control circuit **5022**, a capacitor CT and a comparator **5023**. The charging current source Ic charges the capacitor CT to provide a capacitor voltage Vc. When the capacitor voltage Vc increases to an oscillating reference Vcref, the comparator **5023** flips, and provides the frequency signal Fr with a logic low state. The charge control circuit **5022** receives the capacitor voltage Vc, an upper threshold Vth**1** and a lower threshold Vth**2**, and provides an enable signal EN based thereon. When the capacitor voltage Vc increases to an upper threshold Vth**1**, the enable signal EN enables the discharging current source Id to discharge the capacitor CT.
(40) Then the capacitor voltage Vc decreases. When capacitor voltage Vc decreases to the oscillating reference Vcref, the comparator **5023** flips again, and provides the frequency signal Fr with a logic high state. When the capacitor voltage Vc decreases to a lower threshold Vth**2**, the enable signal EN disables the discharging current source Id. Then the capacitor CT is charged by the charging current source Ic again, the capacitor voltage Vc increases accordingly, and the operation repeats.
(41) In the embodiment of FIG. **5**, the charge control circuit **5022** comprises a first comparator CP**1**, a second comparator CP**2** and a RS flip-flop FF**1**. The first comparator CP**1** receives the capacitor voltage Vc and the upper threshold Vth**1**, and then provides a comparison signal ST to set the RS flip-flop FF**1** when the capacitor voltage Vc increases to the upper threshold Vth**1**. The second comparator CP**2** receives the capacitor voltage Vc and the lower threshold Vth**2**, and then provides a comparison signal RT to reset the RS flip-flop FF**1** when the capacitor voltage Vc decreases to the lower threshold Vth**2**. The RS flip-flop FF**1** provides the enable signal EN to enable the discharging current source Id when be set, and to disable the discharging current source Id when be reset.
(42) The charging current source Ic and the discharging current source Id are controlled by a

current I**1** in the current control circuit **502**A. As shown in FIG. **5**, the current control circuit **502**A includes a frequency control circuit FCT, and a resistor Rf**1**. The frequency control circuit FCT receives the load information signal Sout, and provides a current Ib flowing through the resistor Rf**1** and the frequency control circuit FCT based thereon. Thus the current control circuit **502**A has the current I**1** which is a sum of the current Ia and the current Ib, i.e., I**1**=Ia+Ib. In FIG. **5**, the current Ia flowing through the minimum frequency selecting circuit **501** is Ia=Vdd/Rfmin, wherein Vdd is a power supply voltage provided to the minimum frequency selecting circuit **501**, and Rfmin represents the equivalent resistance of the minimum frequency selecting circuit **501**. The current Ib is determined by the load information signal Sout.
(43) In one embodiment, the charging current source Ic is controlled by the current I**1** to provide a charging current equals to I**1**, and the discharging current source Id is controlled by the current I**1** to provide a discharging current equal to 2×I**1**. In that case, an increasing slope and a decreasing slope of the capacitor voltage Vc are equal, which is I**1**/CT, wherein CT also represents a capacitance of the capacitor CT. The value of the oscillating reference Vcref determines a duty cycle of the frequency signal Fr, while the current I**1** determines a frequency of the frequency signal Fr when the upper threshold Vth**1** and the lower threshold Vth**2** are fixed.
(44) When the frequency generating circuit **50** is utilized with the power systems of the present invention, like the power system **30** in FIG. **3**, the frequency signal Fr is applied to control the frequency of the drive control signal VG, thus to make the high side power switch P**1** and the low side power switch P**2** switch with a frequency determined by the frequency signal Fr. The frequency of frequency signal Fr is regulated by the current Ib which is determined by the load information signal Sout under a normal working condition, wherein the normal working condition refers to a working state that the associated resonant converter working with a frequency above the minimum switching frequency, and a minimum frequency working condition refers to a working state that the associated resonant converter working with the minimum switching frequency. Under the minimum frequency working condition, the current Ib is controlled by the frequency control circuit FCT to be zero, which leads to I**1**=Ia. As a result, the frequency of the frequency signal Fr is corresponds to the current Ia. As can be seen from FIG. **5**, when the power supply voltage Vdd is fixed, the current Ia is determined by the equivalent resistance Rfmin of the minimum frequency selecting circuit **501**, which is further determined by the selecting signal J**1**. That is to say, the minimum frequency of the frequency signal Fr is selected by the selecting signal J**1**.
(45) It should be understood that the frequency generating circuit **50** in FIG. **5** is an example for illustrating how the selecting signal J**1** controls the minimum frequency of the frequency signal Fr, i.e., the minimum frequency of the resonant converter. Many modifications and variations could be adopted. For example, the current I**1** could regulate the upper threshold Vth**1**, the lower threshold Vth**2** or the capacitance of the capacitor CT instead of the current of the charging current source Ic and the discharging current source Id. Furthermore, the current of the charging current source Ic and the discharging current source Id could have other relationships in other embodiments. Also, other oscillating circuit may be utilized to provide a frequency signal based on a current control signal.
(46) Furthermore, as illustrated hereinbefore, the control unit and the minimum frequency selecting circuit may be implemented by digital circuits. In that case, the minimum frequencies are a plurality of frequency values stored in registers or in a lookup table, and could be selected by the selecting signal J**1**.
(47) FIG. **6** schematically shows a power system **60** in accordance with an embodiment of the present invention. As shown in FIG. **6**, the power system **60** includes the power source **301**, the resonant converter **302**, a minimum frequency circuit **604**, the control unit **305** and the drive circuit **306** which may be independent or may represent any combination of one or more components that provide the functionality of the power system **60** as described herein.
(48) The minimum frequency circuit **604** includes: an output voltage detecting circuit **6041**, the

selecting signal generating circuit **3042** and the minimum frequency selecting circuit **3043**. The output voltage detecting circuit **6041** has an input terminal coupled to the output terminal of the resonant converter **302** to detect the output voltage Vout of the resonant converter **302**, and an output terminal configured to provide the feedback voltage Vfb indicative of the output voltage Vout. Compared with the embodiment in FIG. **3**, the feedback voltage Vfb indicative of the output voltage Vout, instead of the equivalent resistance signal Rfb, is provided to the selecting signal generating circuit **3042**. The feedback voltage Vfb is compared with at least one voltage threshold signals to generate the selecting signal J**1**. The rest of the power system **60** works similarly with the power system **30**, and is not described here for brevity.
(49) In one embodiment, the feedback voltage Vfb is compared with a voltage threshold Vref. When the feedback voltage Vfb is larger than the voltage threshold Vref, the minimum frequency signal fmin associated with a lower frequency is provided to the control unit **305**, otherwise, a minimum frequency signal fmin associated with a higher frequency is provided to the control unit **305**.
(50) The output voltage detecting circuit **6041** may include any prior art voltage sense circuit and isolating feedback circuit, like a transformer or an optocoupler.
(51) FIG. **7** shows a control method **70** for controlling a resonant converter in accordance with an embodiment of the present invention. The resonant converter may include any type of LLC converter, LCC converter, CLLC converter, CLLLC converter or the like. The resonant converter works with a frequency and provides an output voltage and an output current to a load. The control method **70** comprises: step **701**, presetting at least two minimum values for a lower limit of the frequency of the resonant converter; and step **702**, selecting one of the at least two minimum values as the lower limit of the frequency of the resonant converter based on the output voltage of the resonant converter or based on an equivalent resistance of the resonant converter, wherein the equivalent resistance is obtained by dividing the output voltage by the output current.
(52) The at least two minimum values of the lower limit of the frequency, i.e., the minimum frequencies are preset according to the specs of the application. For example, persons of ordinary skill in the art could set a frequency corresponds to the peak of the gain curve, or a frequency on the right of the peak (as shown in FIG. **1**) of the gain curve, as the minimum frequency of the corresponding gain curve. Then these minimum frequencies are divided to several groups according to different output voltage ranges, different equivalent resistance ranges, or other specs of the resonant converter. The largest minimum frequency in the group is preset as the minimum frequency for the associated output voltage range or the associated equivalent resistance range. As a result, by the selecting signal J**1**, the minimum frequency is selected for corresponding equivalent resistance or the corresponding output voltage of the resonant converter.
(53) It should be known that presetting the minimum frequencies includes an iterative process, and requires simulation verification for each iteration due to high nonlinearity of the resonant converters.
(54) Obviously, many modifications and variations of the present invention are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims the invention may be practiced otherwise than as specifically described. It should be understood, of course, the foregoing disclosure relates only to a preferred embodiment (or embodiments) of the invention and that numerous modifications may be made therein without departing from the spirit and the scope of the invention as set forth in the appended claims. Various modifications are contemplated, and they obviously will be resorted to by those skilled in the art without departing from the spirit and the scope of the invention as hereinafter defined by the appended claims as only a preferred embodiment(s) thereof has been disclosed.

Claims

1. A power system, comprising: a resonant converter, configured to convert an input voltage to an output voltage to power a load; a control circuit, configured to provide a drive control signal for controlling the resonant converter working with a frequency having a minimum value varied based on the output voltage; and wherein the control circuit comprises: an output voltage detecting circuit, configured to provide a feedback voltage based on the output voltage; a selecting signal generating circuit, configured to provide a selecting signal based on the feedback voltage; and a minimum frequency selecting circuit, configured to provide a minimum frequency signal based on the selecting signal for determining the minimum value of the frequency of the resonant converter, wherein the minimum frequency signal associated with a lower frequency is provided when the selecting signal indicates the feedback voltage is larger than a voltage threshold, and the minimum frequency signal associated with a higher frequency is provided when the selecting signal indicates the feedback voltage is less than the voltage threshold.
2. The power system of claim 1, wherein the output voltage detecting circuit comprises a transformer.
3. The power system of claim 1, wherein the output voltage detecting circuit comprises an optocoupler.
4. The power system of claim 1, wherein the minimum frequency selecting circuit comprises a look-up table providing the minimum frequency signal based on the selecting signal.
5. The power system of claim 1, wherein the control circuit further comprises a current-frequency converting circuit, and wherein: the minimum frequency selecting circuit comprises a resistor matrix with a resistance determined by the selecting signal, and the current-frequency converting circuit provides the drive control signal with a minimum frequency determined by a current flowing through the resistor matrix.
6. A control circuit of a resonant converter, wherein the resonant converter provides an output voltage to power a load, comprising: a minimum frequency circuit, configured to provide a minimum frequency signal based on the output voltage of the resonant converter; a control unit, configured to provide a drive control signal for controlling the resonant converter working with a frequency having a minimum value determined by the minimum frequency signal; and wherein the minimum frequency circuit comprises: an output voltage detecting circuit, configured to provide a feedback voltage based on the output voltage; a selecting signal generating circuit, configured to provide a selecting signal based on the feedback voltage; and a minimum frequency selecting circuit, configured to provide the minimum frequency signal based on the selecting signal, wherein the minimum frequency signal determines a lower frequency when the selecting signal indicates the feedback voltage is larger than a voltage threshold, and the minimum frequency signal determines a higher frequency when the selecting signal indicates the feedback voltage is less than the voltage threshold.
7. The control circuit of claim 6, wherein the minimum frequency selecting circuit comprises a look-up table providing the minimum frequency signal based on the selecting signal.
8. The control circuit of claim 6, wherein the control unit comprises a current-frequency converting circuit, and wherein: the minimum frequency selecting circuit comprises a resistor matrix with a resistance determined by the selecting signal, and the current-frequency converting circuit provides the drive control signal with a minimum frequency determined by a current flowing through the resistor matrix.
9. A control circuit of a resonant converter, wherein the resonant converter provides an output voltage and an output current to power a load, comprising: a minimum frequency circuit, configured to provide a minimum frequency signal based on the output voltage of the resonant converter; and a control unit, configured to provide a drive control signal for controlling the resonant converter working with a frequency having a minimum value determined by the minimum frequency signal; and wherein the minimum frequency circuit comprises: an equivalent resistance

detecting circuit, configured to provide an equivalent resistance signal based on the output voltage and the output current; a selecting signal generating circuit, configured to provide a selecting signal by comparing the equivalent resistance signal with a resistance threshold signal; and a minimum frequency selecting circuit, configured to provide the minimum frequency signal based on the selecting signal, wherein a first minimum frequency is determined by the minimum frequency signal when the selecting signal indicates the equivalent resistance signal is smaller than the resistance threshold signal, and a second minimum frequency is determined by the minimum frequency signal when the selecting signal indicates the equivalent resistance signal is larger than the resistance threshold signal, and the second minimum frequency is less than the first minimum frequency.

10. The control circuit of claim 9, wherein the minimum frequency selecting circuit comprises a look-up table providing the minimum frequency signal based on the selecting signal.

11. The control circuit of claim 9, wherein the control unit comprises a current-frequency converting circuit, and wherein: the minimum frequency selecting circuit comprises a resistor matrix with a resistance determined by the selecting signal; and the current-frequency converting circuit provides the drive control signal with a minimum frequency determined by a current flowing through the resistor matrix.